# MINI

## 384×288/640×512

## Uncooled Infrared Thermal Module

## Product Manual V1.10

# Content

# Change History

Changes between document issues are cumulative. The latest document issue contains all the changes in previous issues.

| Version | Release Date | Revision Content |
|---|---|---|
| **V0.1** | July 30, 2020 | Create |
| **V0.2** | October 30, 2020 | Updated power consumption parameters; Added SPI interface information; Updated the picture of the interface board. |
| **V0.3** | December 22, 2020 | Added adaptive lens parameters; Updated the temperature measurement accuracy information; Updated power consumption parameters. |
| **V0.4** | June 1st, 2021 | Updated product pictures, product features, product model selection table, lens parameters, power consumption parameters, communication interface information, temperature measurement range and accuracy and mechanical drawings. Added connector model number. |
| **V1.0** | August 31, 2021 | Updated the serial communication interface information; Update product performance parameters and comments; Add interface board pin definition; New data output and control; Delete digital video information; Add power-on sequence diagram, power-off sequence diagram and forced update sequence diagram. |
| **V1.1** | October 14, 2021 | Updated lens specifications; Update module power consumption; Update user interface component product legend and pin definition legend; Update that product pin definition of the user interface component; Add the definition and diagram of the pins of the supporting tooling line; Update DVP timing; Update the caption. |
| **V1.2** | December 9, 2021 | Added comments on product performance temperature measurement parameters; Update DVP timing; Update the output information format; |
| **V1.3** | December 30, 2021 | Added the definition of analog video in DF40C-50DP-0.4V(51); Added I2C control description in WN00-V101F001C. |
| **V1.4** | April 6, 2022 | Re-planned the product selection; The temperature parameter range is updated; Optimization simplifies the user's instructions. |

| Version | Release Date | Revision Content |
| --- | --- | --- |
| **V1.5** | March 3, 2023 | Adjusted the layout and chapter structure of the manual; Increase optional models and lens selection; Update USB expansion component information; Update SDK usage library; Update the temperature parameter range. |
| **V1.6** | October 18, 2023 | 1. Updated product selection.<br>2. Updated specifications: updated detector framerate and video output format instructions.<br>3. Update the pin definition description table: delete DATA0~DATA7.<br>4. Added warranty instructions.<br>5. Update the style. |
| **V1.7** | February 2, 2024 | Updated product and lens model. |
| **V1.8** | May 16, 2024 | Updated the document style. |
| **V1.9** | May 22, 2024 | Updated the document style. |
| **V1.10** | June 14, 2024 | Updated structural drawings. |

# 1. Product Introduction

## 1.1. Product Description

MINI-384/640 series uncooled infrared thermal module adopts processing chip independently developed by our company to replace the traditional FPGA scheme of thermal imaging module. In addition, WN-384/640 series adopts a new self-developed 12 μm pixel WLP package detector, which provides a parallel digital interface for easy integration and development and can be flexibly connected to various intelligent processing platforms. The module has the characteristics of high performance, small volume, light weight, low power consumption and low cost, and meets the application requirements of SWaP (size, weight, and power/price).

Figure 1-1 MININ-384/640 Module

## 1.2. Product Features

- Compact in size.
- Super low power consumption.
- High quality image display.
- High-quality/wide-range configurable temperature measurement.

## 1.3. Application Scenarios

- Smart life: smart home appliances, smart sensors.
- Handheld terminal: built-in or external plug-in for mobile phone, temperature measuring tool, night vision equipment.

- Security inspection: industrial monitoring, perimeter scanning, inspection robot, photoelectric pod.
- Fire safety: fire warning, fire helmet.
- Medical health: medical heat map, cell temperature measurement.

# 2. Product Selection

## 2.1. Model Introduction

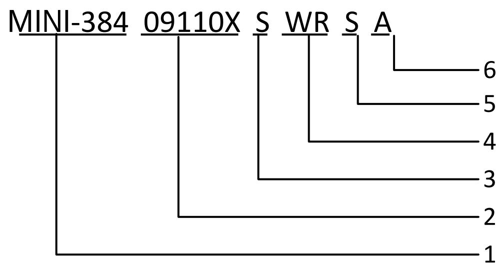


Table 2-1 MINI-384/640 Product Model Number Explanation

| No. | Name | Code | Description |
|---|---|---|---|
| **1** | Name | MINI-384 | Resolution: 384 * 288 |
| | | MINI-640 | Resolution: 640*512 |
| **2** | Lens (focal length, F-number) | 3D911X | 3.9 mm F1.1 |
| | | 06812X | 6.8 mm F1.2 |
| | | 09110X | 9.1 mm F1.0 |
| | | 13510X | 13.5 mm F1.0 |
| | | 18011X | 18 mm F1.1 |

| No. | Name | Code | Description |
|---|---|---|---|
| **3** | Gain mode | H | High-quality |
| | | S | High-quality/ Wide range (configurable) |
| **4** | Thermal Core Mode | WR | Temperature Measurement |
| **5** | Focus Type | S | Fixed focus |
| | | M | Manual focus |
| **6** | Lens Serial Number | A | The first lens for the focal length, F-number and focus method. |
| | | B | The second lens. |

NOTE 1.High-quality mode refers to the high-gain mode, where the module response rate is high, and the imaging quality is good, but the temperature measurement range is narrow; Wide-range mode refers to the original low gain mode, where the module response rate is low, and the imaging quality is decreased, but the temperature measurement range is wide.

## 2.2. Recommended Models

Table 2-2 List of recommended models for WN-384/640

| Lens | WN-384 | WN-640 |
|---|---|---|
| **f3.9mm F1.1** | MINI384_3D911X SWR MA | × |
| **f6.8mm F1.2** | MINI384_06812X SWR MA | × |
| **f9.1mm F1.0** | MINI384_09110X SWR MA | MINI640_09110X SWR SA |
| **f13.5mm F1.0** | MINI384_13510X SWR MA | MINI640_13510X SWR SA |

| Lens | WN-384 | WN-640 |
|---|---|---|
| **f18mm F1.1** | × | MINI640_01811X SWR MA |

NOTE 1.The symbol "×" indicates that the model is not supported.

NOTE 2.If necessary, you can contact our company to customize the selection combination that is not recommended in the list.

## 2.3. Lens Selection Reference

Table 2-3 MINI-384 Lens Parameters

| Lens Code | Lens Parameters | | | | Focus | Detection Distance | Recognition Distance | Identification Distance | Temperature Measurement Distance | Weight (Lens and Flange) |
|---|---|---|---|---|---|---|---|---|---|---|
| | Focal Length | F-number | FOV | IFOV | | | | | | |
| **3D911X** | 3.9 mm | 1.1 | 65°×50° | 3.07 mrad | 0.3 m~∞ | 0.43 km | 0.1 km | 0.054 km | 0~8 m | 12g±2% |
| **06812X** | 6.8 mm | 1.2 | 40°×30° | 1.76 mrad | 0.8 m~∞ | 0.76 km | 0.19 km | 0.09 km | 0~10 m | 12g±5% |
| **09110X** | 9.1 mm | 1.0 | 29.2°×21.7° | 1.3 mrad | 1.69 m~∞ | 1.1 km | 0.25 km | 0.17 km | 0~20 m | 10.5g±5% |
| **13510X** | 13.5 mm | 1.0 | 19.6°×14.7° | 0.9 mrad | 3.31 m~∞ | 1.5 km | 0.37 km | 0.19 km | 0~20 m | 15g±5% |

Table 2-4 MINI-640 Lens Parameters

| Lens Code | Lens Parameters | | | | Focus | Detection Distance | Recognition Distance | Identification Distance | Temperature Measurement Distance | Weight (Lens and Flange) |
|---|---|---|---|---|---|---|---|---|---|---|
| | Focal Length | F-number | FOV | IFOV | | | | | | |
| **09110X** | 9.1 mm | 1.0 | 48.7°×38.6° | 1.3 mrad | 1.69 m~∞ | 1.1 km | 0.25 km | 0.17 km | 0~20 m | 10.5g±5% |
| **13510X** | 13.5 mm | 1.0 | 31.9°×25.7° | 0.9 mrad | 3.31 m~∞ | 1.5 km | 0.37 km | 0.19 km | 0~20 m | 15.2g±5% |
| **18011X** | 18 mm | 1.1 | 24.2°×19.5° | 0.7 mrad | 6.07 m~∞ | 2 km | 0.5 km | 0.25 km | 0~20 m | 16g±5% |

NOTE 1. The above detection distance, recognition distance and identification distance are estimated according to Johnson's Criteria, taking pedestrians (1.8×0.5×0.3 m) as the target. The temperature measuring distance is estimated by a non-point source blackbody with a diameter of 0.15 m.

NOTE 2. The temperature correction here refers to the range of distance correction tables, which can be provided. Users can correct the accuracy of temperature measurement targets at a longer distance by other distance correction methods.

NOTE 3. The weight here only indicates the weight of the lens. The whole weight includes lens, module, and the USB accessories. The weight of the USB accessories is 2.8g±5%. The module weight (lens and USB accessories excluded) is 6.3g±5%.

NOTE 4. Focus refers to the maximum clear range that can be obtained when fixed focusing (dispensing or fixed using a headless set screw), not the focusing range, which is typically 0.3 m~∞.

# 3. Specifications

Table 3-1 MINI-384/640 System Specifications*1*2

| Overview | MINI-640 | MINI-384 |
|---|---|---|
| **Detector Type** | Vanadium Oxide Uncooled Infrared Focal Plane Detector | |
| **IR Wavelength** | 8～14μm | |
| **Resolution** | 640×512 | 384×288 |
| **Pixel Pitch** | 12μm | |
| **Noise Equivalent Temperature Difference (NETD)** | ≤50mK@25℃,F#1.0 (≤40mK optional) | |
| **Thermal Time Constant** | ＜12ms | |
| **Detector Framerate*3** | Temperature measurement module: 25Hz; Imaging module: 50Hz | |
| **Non-uniformity Correction** | Auto Shutter Correction | |
| **Image Output Formats** | 10bit/14bit (configurable) | |
| **Focusing Mode** | Fixed focus/Manual focus | |
| **Image Adjustment** | | |
| **Brightness** | 0~255 (configurable) | |
| **Contrast** | 0~255 (configurable) | |
| **Polarity** | White-hot/Black-hot | |
| **Palette** | Supported | |
| **Mirroring** | Vertical/Horizontal/Diagonal | |
| **Image Processing** | TEC-less Algorithm | |
| | Non-uniformity Correction | |
| | Digital Filtering Noise Reduction | |
| | Digital Detail Enhancement | |
| **Electricity** | | |
| **Input Supply Voltage** | 3.3V | |
| **Video Output Format** | DVP, BT656, USB, CVBS (white-hot) | |
| **Command and Control Interface** | I2C | |
| **Typical Power Consumption @25℃** | Normal: <0.55W | Normal: <0.50W |

| | | |
|---|---|---|
| | Peak: <1.23W | Peak: <1.18W |
| **Physical Characteristics of Module (lens and flange not included)** | | |
| **Package Dimension (W×L×H)** | 21mm×21mm×8mm | |
| **Weight** | <8 g | |
| **Thermography*3** | | |
| **Measurement Range** | High-quality: -20℃ ~ +150℃; Wide-range: 0℃ ~ +550℃ | |
| **Measurement Accuracy (typ.)** | Accuracy for range -20℃~+150℃: ±2℃ or ±2% (choose the greater value)， @-20℃~60℃ ambient temperature<br>Accuracy for range 0℃~+550℃: ±3℃ or ±3% (choose the greater value)， @-20℃~60℃ ambient temperature | |
| **Environment** | | |
| **Operating Temperature** | -40℃~+80℃ | |
| **Storage Temperature** | -50℃~+85℃ | |
| **Humidity** | 5%–95%, non-condensing | |
| **Vibration** | 6.06g, random vibration, all axes | |
| **Impact** | 80g@4ms, final peak sawtooth wave, three axes and six directions | |

NOTE 1. The above parameters are measured under laboratory conditions, the actual specifications and test methods are subject to the test report.

NOTE 2. For applications requiring high temperature measurement accuracy and image uniformity, the module needs a secondary temperature correction and lid-pattern noise correction after installation. Please refer to the Quick Start Manual for details.

NOTE 3. The imaging module does not include the temperature measurement function.

# 4. Hardware Introduction

## 4.1. Pin Definition

50PIN connector model number: DF40C-50DP-0.4V（51）

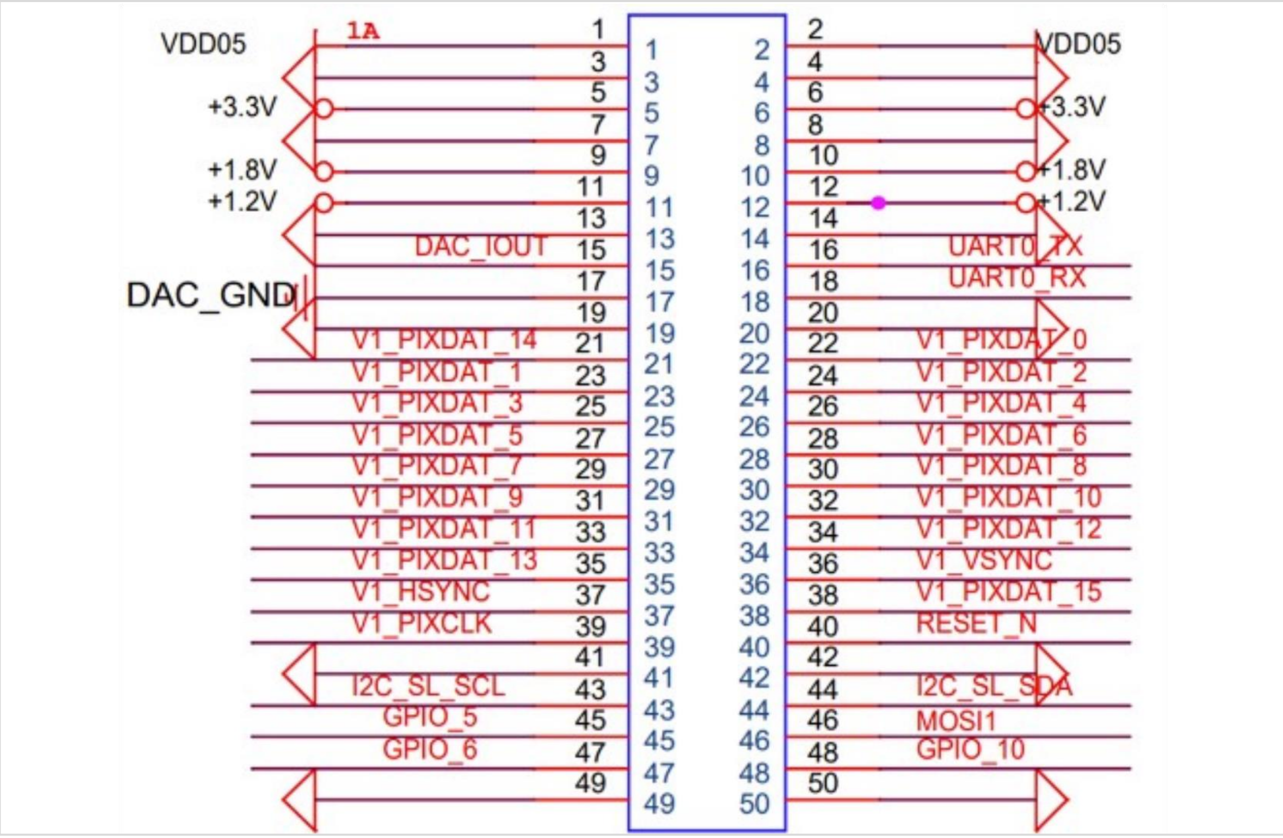


Figure 4-1 Hirose50 Pinout Diagram A

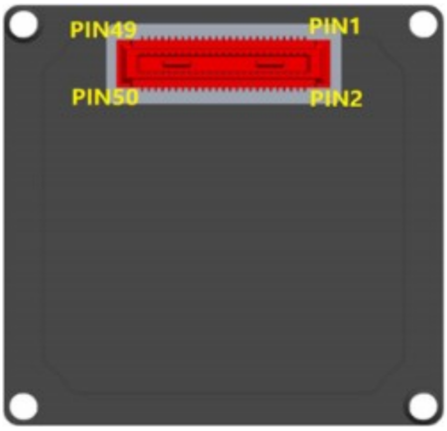


Figure 4-2 Hirose50 Pinout Diagram B

Table 4-1 HIROSE50 PIN Definition

| No. | Name | Type | Revision Content | |
|---|---|---|---|---|
| **1, 2** | VDD5.0 | Power | External power input 5V[1] | |
| **5, 6** | VDD3.3 | Power | External power input 3.3V[2] | |
| **9, 10** | VDD1.8 | Power | External power input 1.8V[3] | |
| **15** | DAC_IOUT | Output | Analog video | |
| **17** | DAC_GND | Output | Analog video GND | |
| **21** | DATA14 | OUT | 16Bit/8Bit parallel data | Data signal |
| **30** | DATA8 | | | Data signal LSB (8bit) |
| **31** | DATA9 | | | Data signal |
| **32** | DATA10 | | | Data signal |
| **33** | DATA11 | | | Data signal |
| **34** | DATA12 | | | Data signal |
| **35** | DATA13 | | | Data signal |
| **36** | VSYNC | | | Field synchronization signal |
| **37** | HSYNC | | | Line synchronization signal |
| **38** | DATA15 | | | Data signal MSB |
| **39** | CLK | | | Clock signal |
| **40** | RESET | IN | Reset | |
| **11, 12, 16, 18, 45~48** | NA | NA | Not available | |
| **43** | SCL | OUT | I2C clock line | |
| **44** | SDA | I/O | I2C data line | |
| **3, 4, 7, 8, 13, 14, 19, 20, 41, 42, 49~50** | GND | Power | Power ground | |

NOTE 1. The hardware description here is only for product selection reference. Please contact our technicians for detailed hardware development information during specific development.

Power Supply Requirement:

1) The typical power input voltage is 5 VDC, where the voltage value refers to the voltage to the module connector. The power rising time (10%~90%) is <4 mS; the peak current <1.0 A; and the ripple and noise <40 mVp-p.

2) The typical power input voltage is 3.3 VDC, where the voltage value refers to the voltage to the module connector. The power rising time (10%~90%) is <4 mS; the peak current <0.6 A; and the ripple and noise <40 mVp-p.

3) The typical power input voltage is 1.8 VDC (1.75V<X<1.85V), where the voltage value refers to the voltage to the module connector.

4) DVP, BT656 only support 8bit output, CVBS only support white-hot (switching black-hot can be customized).

# 5. Functions

Modules are provided with SDKs, demonstration software, and their adapted drivers.

## 5.1. Software Resources

### 5.1.1. SDK

The SDK provides graphing and control interfaces and infrared temperature algorithms. Windows and Linux (including embedded systems) versions are supported. The following libraries are provided: libiruvc, libirtemp, libirparse, libirprocess, libircmd, libiri2c, and the sample project libirsample which calls the above six libraries, please refer to the SDK documentation for detailed description.

### 5.1.2. Demonstration Software

Thermal Cam is the upper computer software used to demonstrate the effect of the module, based on SDK library and QT development, support Windows version. It connects via USB and has functions such as image display, adjusting image parameters, temperature measurement, switching palette mode and so on.

## 5.2. Sequencing and Control

### 5.2.1. Data Format

Customers can select three output mode: Image Mode, Temperature Mode and Image plus Temperature Mode. The Image Mode supports displaying both grayscale and color images. In addition, NUC and AGC data can be obtained.

Table 5-1 Data Format Description

| Mode | Data Type | Data Format |
|---|---|---|
| Image | Grayscale images | Y14/Y10 |
| | Color images | YUV422 |
| Temperature | Temperature data | Y14 |
| Image + Temperature | Image + Temperature | All |

### 5.2.2. Interfaces

Includes control interfaces and video interfaces.

**CCI Control Interface**

The CCI control interface adopts standard I2C. The maximum clock frequency is 400KHz, complies with the CCI protocol in MIPI CSI-2.

**CCI Video Interface: DVP**

The DVP interface data is transmied via single channal (8bit), i.e. DATA8~DATA15.

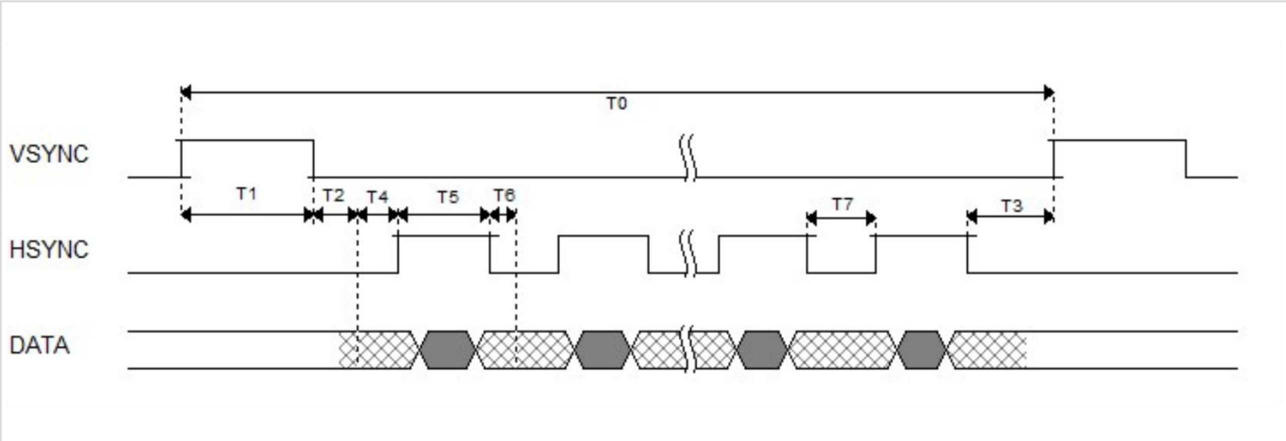


Figure 5-1 Single Channel Sequencing Diagram A

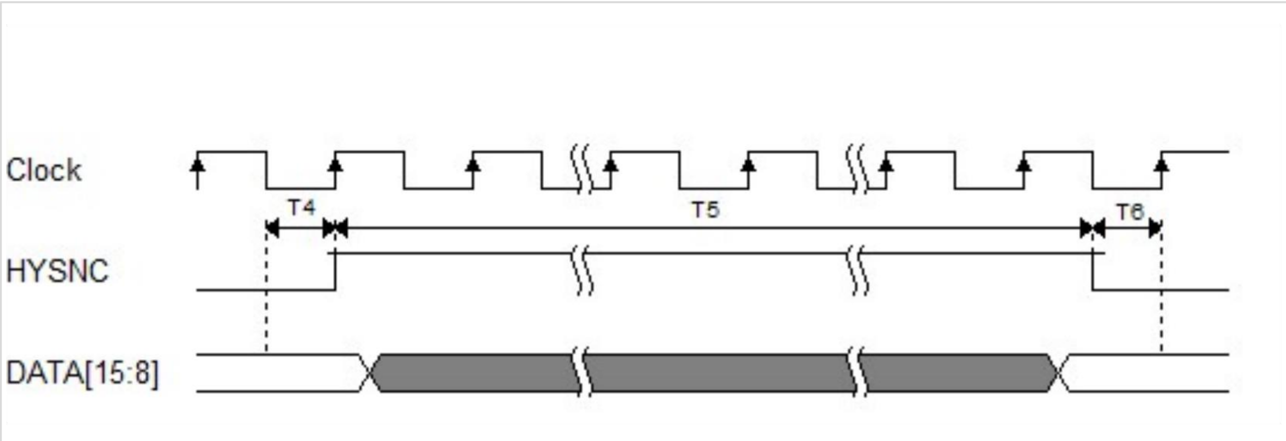


Figure 5-2 Single Channel Sequencing Diagram B

**Special Cases**

This section is applicable to situations where the hardware requirements of the client directly control the power on, off, and forced updates of the module.

1. Power-on sequence diagram

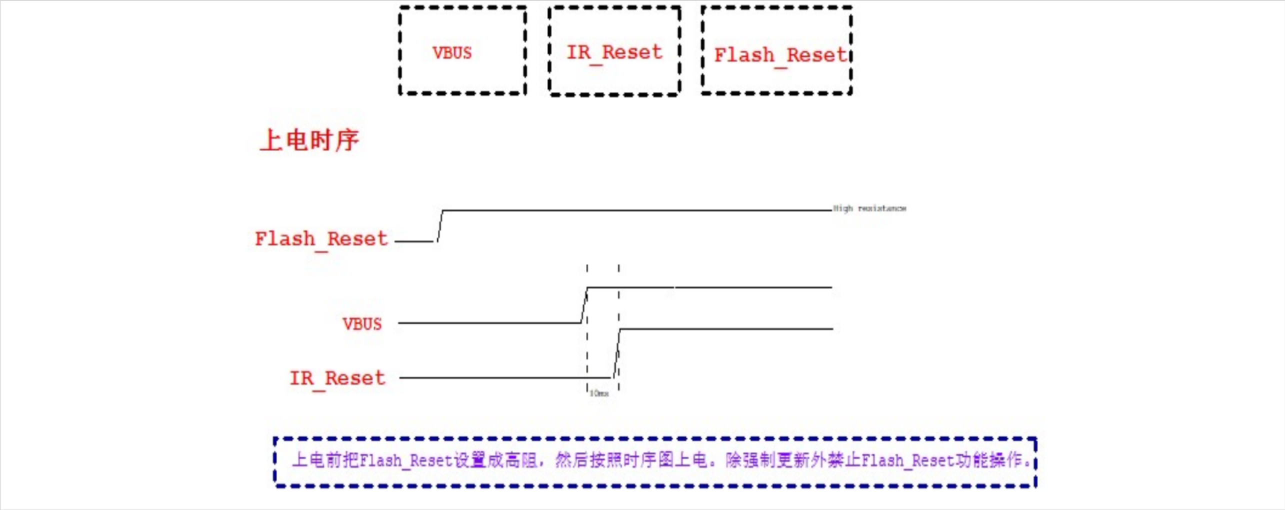


Figure 5-3 Power-on Sequence Diagram

2. Power-off sequence diagram

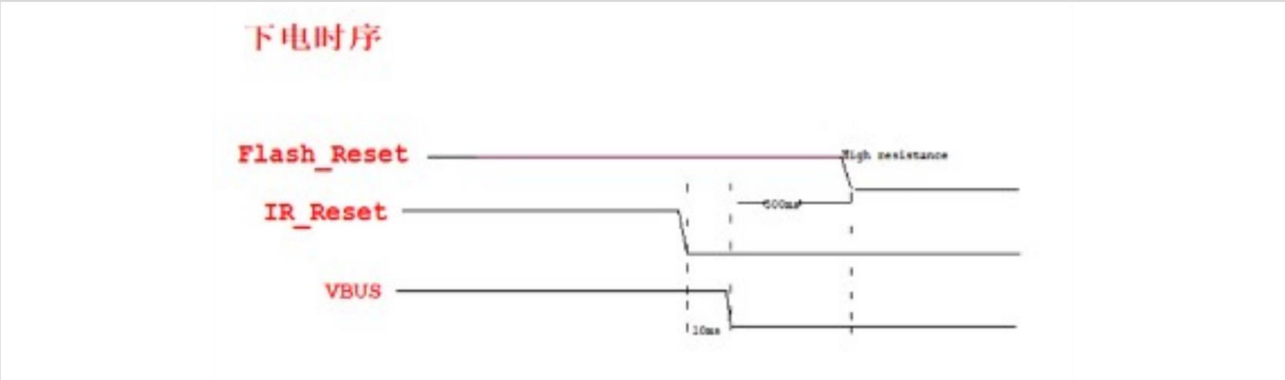


Figure 5-4 Power-off Sequence Diagram

3. Forced update sequence diagram

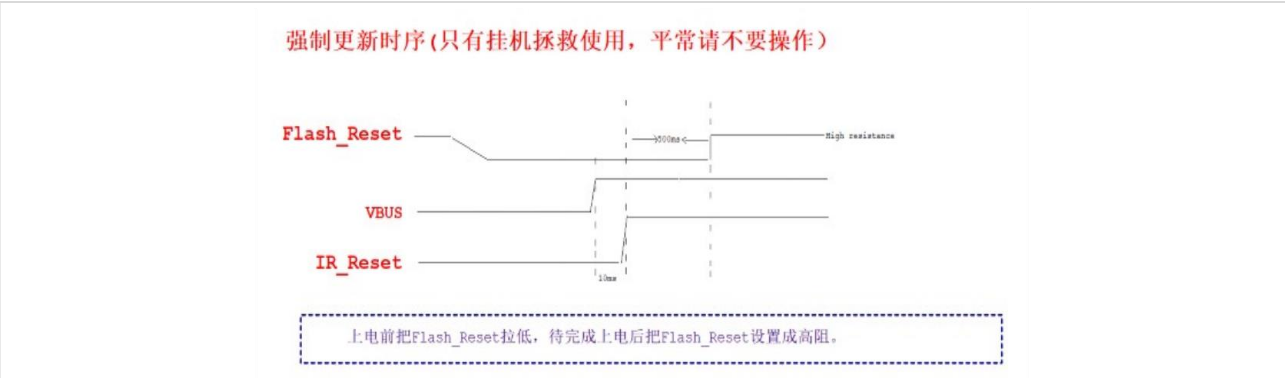


Figure 5-5 Forced Update Sequence Diagram

## 5.3. Module Functions

Table 5-2 MINI-384/640 Functions

| **Image-Related** | **Adjustable Effect** | **Remarks** |
|---|---|---|
| **Palette** | 12 modes in total | One is reserved |
| **Flip** | Supported | |
| **Digital Noise Reduction** | Temporal Noise Reduction and Spatial Noise Reduction | |
| **Automatic Image Adjustment** | Automatically adjust the image stretching ratio according to the scene to make the brightness and contrast of the image within the appropriate range. | |
| **Digital Detail Enhancement** | Make image edges, details, etc. clearer | |
| **Supplemental Flat Field Correction** | Eliminate the non-uniformity inherent in the optical system. | |
| **Dead Pixel Correction** | Eliminate a single dead pixel in the picture. | |
| **Temperature Measurement** | **Adjustable Effect** | **Remarks** |
| **Obtain Detector Temperature** | Determine whether the module is thermally stable and provide reference for subsequent temperature correction. | Not available |
| **Temperature Measurement Analysis** | Point-line-rectangle, max/min temperature measurement | More functions can be realized on the software side after acquiring the whole frame temperature data. |
| **Temperature Measurement Alarm*** | Remind when the set temperature threshold is exceeded. | |

| | | |
|---|---|---|
| **Ambient Variables Correction*** | Ambient reflection temperature, atmospheric temperature, target emissivity and atmospheric transmittance (distance correction) | |
| **Secondary Calibration for Temperature Measurement** | Recalibrate the temperature deviation caused by optical changes or dissipation structure changes. | |
| **Temperature Measurement Method** | Points, lines, rectangles, entire frames | |
| **Correction** | Distance, reflectivity | |
| **Shutter-Related** | **Available Functions** | **Remarks** |
| **Manual Shutter Correction** | Shutter correction makes the image more even and the temperature measurement more accurate. | |
| **Automatic Shutter Setting** | Includes automatic shutter switch, time interval setting and ambient temperature interval setting. | |
| **Other Sections** | **Available Functions** | **Remarks** |
| **Obtain Product Information** | Read data such as module SN, PN, version number. For easier product identification and maintenance. | |
| **Read/Write Customized Filed** | Read/Write customized address and length. Make the product easy to be recognized by software. | Only for specified modules |
| **Firmware Upgrade** | Push a new firmware version to the module. | Only support version upgrade after ver.3.1. |
| **Configuration Switch** | Switch between high-quality and wide-range modes. | |

NOTE: Function with * only supported by SDK.

# 6. Structure and Dimensions

NOTE: The drawing below includes 9.1 mm lens module. If you need drawings for other products models, please contact our technicians for further information.

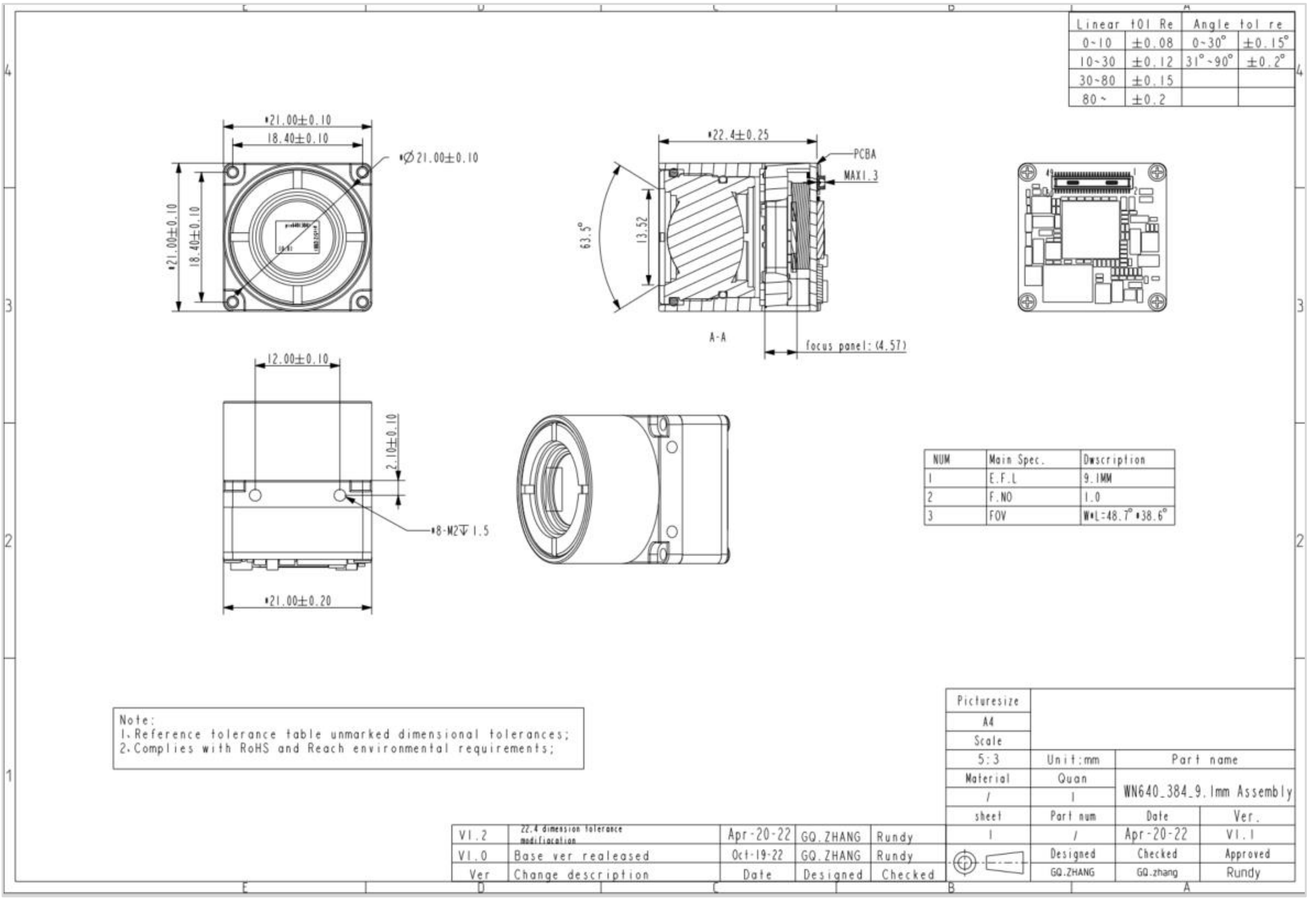


Figure 6-1 MINI-384/640 Mechanical Drawings

# 7. Accessories

## 7.1. USB Kit

The 4-PIN connector model number is: FWFS-08R1-04-19.


Figure 7-1 USB Kit

Table 7-1 Connector PIN Definition

| No. | Name | Type | Revision Content |
|---|---|---|---|
| **1** | VBUS_IN | Power | External power input: 5V |
| **2** | GND | GND | GND |
| **3** | USB_DP | I/O | USB2.0 Signal |
| **4** | USB_DM | I/O | USB2.0 Signal |

# 8. Precautions

To protect you and others from injury or to protect your equipment from damage, please read all the information below before using the equipment.

1. Do not look directly at high-intensity radiation sources such as the sun.
2. The ideal operating environment temperature is -40℃~80℃.
3. Do not touch devices and cables with wet hands.
4. Do not bend or damage the cables.
5. Do not scrub your device with diluent.
6. Do not unplug other cables before cutting off the power supply.
7. Do not connect the attached cables in the wrong way to avoid damaging the device.
8. Take measures to prevent static electricity.
9. Do not dismantle the device. If there is any fault, please contact us to have it repaired by professionals.

# 9. Warranty

Dear User,

Thank you for choosing our products, we will, as always, to provide you with satisfactory service!

1. This product in normal use under the circumstances of failure the company will provide 1-year warranty, life-long maintenance services.
2. Warranty scope:

   Failure under normal circumstances is generally defined as natural damage caused by the user of the product during normal use without human intent or due to negligence factors.
3. The following cases are not in our warranty scope:
   1) Any damage caused by modification or repair not authorized and permitted by the company.
   2) Failure or damage caused by the use of third-party product software, service behavior.
   3) Accidental factors or human behavior caused by product damage. Such as into the liquid, drop damage, input the wrong voltage, excessive extrusion, deformation of the motherboard and so on. Appearance of obvious hard object damage, cracks, broken foot, serious deformation, power cord broken, broken wire, bare core and other phenomena.
   4) Product data loss or damage.
   5) Cannot effectively present the product warranty certificate. (Product nameplates, SN barcodes, and tamper-evident labels are torn off or damaged, blurred and unrecognizable.)
   6) Not in accordance with the instructions for installation, use, maintenance, storage of the product failure or damage.
   7) It has exceeded the warranty period.
   8) Failure or damage due to uncontrollable factors (e.g. fire, earthquake, flood, etc.).

# 10.Supports and Services

The company provides services such as pre-sales training, in-sales support, and after-sales maintenance. Please contact our sales-reps for specific information.

# 11. contacts

E-mail:

sale1@liberal-technology.com

sale2@liberal-technology.com

sale3@liberal-technology.com